Ulfatullah Sk

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I am a GIS Proffesional with 2.8 years of experience. I have worked with multiple technologies over time. Below is a general overview of my tech stack, which I have used in nearly every application I have built or worked on.

- **GIS Software & Tools** - ArcGIS, ArcGIS Pro, QGIS, ERDAS IMAGINE, ESA SNAP,Agisoft, ENVI, Google Earth Engine (GEE), etc.
- **DS & GeoPython** - Python, Pandas, NumPy, GDAL, Matplotlib, Seaborn, GeoPandas, Geemap, Leafmap, Shapely, Rasterio, ArcPy, PyQGIS, etc.
- **Frontend** - HTML, CSS, JavaScript, OpenLayers, React.
- **Backend** - Python, Geoserver, FastAPI.
- **Database** - SQL, PostgreSQL, PostGIS.
- **Other** -DSSAT, Canva, Microsoft Office Suite.

Education

Guru Jambheshwar University of Science and Technology, Aug, 2021 - July, 2023
M. Tech In Geo-Informatics
Hisar, India
CGPA: 7.44

Experience

GIS Analyst – Mahyco Pvt. Ltd. July, 2025 - Present
Jalna, India

- **Crop Yield Estimation** – Developed semi-physical crop yield estimation models using satellite, weather, and agronomic data.
- **DSSAT Modeling** – Worked on crop simulation and productivity assessment using DSSAT.
- **Drone Phenotyping** – Processed UAV imagery for crop phenotyping and precision agriculture applications.
- **Remote Sensing Analytics** – Performed satellite-based crop monitoring and time-series analysis using vegetation indices.
- **Web GIS-Based Crop Health Visualization** – Developed interactive GIS-based visualization for monitoring crop health status and Yield.
- **Climate Data Processing -** Processing IMD, NASA POWER, MOSDAC, ERA5, and WRF datasets for crop climate impact and climate variability assessment.
- **GeoAI & Machine Learning** – Building AI/ML-based crop yield estimation models for improved agricultural forecasting and decision support.
- **Process Automation** – Developed automated workflows for geospatial data processing, analysis, and reporting using Python (GDAL) and QPy.
- **Technology Stack** – Python, QPy, Machine Learning, GDAL, DSSAT, Google Earth Engine (GEE), QGIS, UAV Data Processing.

Project Fellow – Haryana Space Application Centre (HARSAC) Feb, 2024 - Jun,2025
Hisar, India

- **Time-Series:** Time-series mapping and classification of Kharif and Rabi crops using multi-temporal satellite imagery.
- **YesTech Project :**Supported crop yield estimation using PAR, fAPAR, and RUE-based semi-physical models; performed QA/QC and output validation.
- **Fasal Project :**Conducted crop monitoring, classification, and geospatial analysis for agricultural assessment.
- **Tools – Arcapy (Process Automation)GEE, ArcGIS Pro, QGIS, Remote Sensing & Geospatial Analysis.**

My Projects

Title - GIS Tools Automation

- **Process Automation** – Automated geoprocessing workflows, spectral indices (NDVI, NDWI, EVI, LST) and terrain analysis (DEM, DSM, DTM, Slope, Aspect, Hillshade).
- **Change Detection & Digitization** – Performed LULC change detection and digitization of buildings, roads, agricultural land, forests, and other features.

Title - Geo Web Platform

- **Web App:** A Web GIS and GeoAI application for satellite data visualization and geospatial analytics. Supports GeoTIFF, WMS, WFS, GeoJSON, and Vector Tiles with FCC, NDVI, NDWI visualization, pixel inspection, feature query, spatial analysis, caching, and dynamic layer management.
- **Features**: Satellite Visualization, Feature Query, Pixel Inspector, Crop Health Monitoring, Time-Series Analysis, Authentication, and Map Export.

Student Trainee– Haryana Space Application Centre (HARSAC) 2021 – 2023

- **Paddy Acreage Estimation** – Estimated paddy acreage using AI/ML-based crop classification models.
- **Model Evaluation** – Assessed and compared multiple machine learning models for crop acreage estimation.
- **Biomass & Yield Assessment** – Estimated paddy biomass and yield in major rice-growing districts of Haryana.
- **Yield Prediction** – Integrated acreage estimation with semi-physical modeling to improve prediction accuracy.

Summer Intern - School of Environmental Sciences, Jawaharlal Nehru University (JNU), New Delhi Jul 2022 – Aug 2022
New Delhi, India

- Analyzed NO_2, SO_2, and temperature variations over Delhi using spatial analysis techniques.
- Applied IDW interpolation for pollutant distribution mapping and environmental assessment.

Languages

- English
- Hindi
- Bengali

Certifications & Training and Publication

- **Advanced Statistical & Machine Learning Techniques for Data Analysis in Agriculture** – ICAR-NIASM
- **Geodata Processing Using Python** – Indian Institute of Remote Sensing (IIRS), ISRO (2023)
- **Advanced Geospatial Technologies for Disaster Risk Reduction** – IIRS, ISRO (2022)
- **21-Day GIS Training Program Using QGIS** – Central University of Karnataka & SIUD (2020)
- **Publication**: Co-Author of Geospatial and Multivariate Analysis of Soil Properties for Delineation of Management Zones in Arid to Semi-Arid Agroecosystems of Southern Haryana, India.
- **Intregrated Capacity Building on Digital Agriculture Cro MOdels, AI, IoT & Risk Mapping.**
- (Unesco Chair & AVV) 2026.